Research Statement

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My primary research interest lies in advancing the computational theory of transformative learning technologies and translating these advances into practical applications that improve learning and teaching while contributing to the broader theory of the learning sciences. I am particularly focused on artificial intelligence technologies that support students in learning, assist teachers in instruction, and enable researchers to better understand how people learn. As such, my scholarly contributions span computer science, cognitive science, and the learning sciences.

To expand the computational foundations of transformative learning technologies, I employ human-centered design and iterative engineering methods. I begin by identifying the needs and challenges of students and teachers, often using user-centered techniques rooted in human-computer interaction research. Based on these insights, I design and develop AI-driven technological solutions. I then rigorously evaluate these solutions in authentic classroom environments through field studies involving real students and teachers, yielding large-scale empirical data.

I analyze these empirical data using both traditional statistical approaches and advanced data mining techniques to develop theoretical models of computing for learning technologies. In parallel, to advance the theory of learning and teaching, I formulate and test specific hypotheses about how people learn and what makes teaching more effective. These investigations often take the form of mixed-method studies, including randomized controlled trials conducted in business-as-usual classroom settings.

My intellectual leadership in this field is evidenced by my scholarly achievements with 32 journal and 59 conference/workshop papers at the top-tier venues. I have also demonstrated my strengths in securing highly competitive funding, including being a two-time awardee of the Institute of Education Sciences (IES) of the U.S. Department of Education—one of the most prestigious and selective research funding programs in education, in addition to eight awards from the National Science Foundation.

1. Research Accomplishments

Although the advanced technologies I develop and apply in my research are generally domain-independent, my primary focus has been on mathematics education—an area grounded in my academic background, with both my bachelor's and master's degrees concentrated in math education. Over the past 20 years, I have built a sustained and impactful research program in this domain. The following sections outline my key research accomplishments and contributions during this period.

1.1 Studying a Computational Theory of Learning and Its Applications

I am perhaps best known for my work on **SimStudent** (www.SimStudent.org), a long-standing research program centered on developing and studying a computational model of learning. At the behavioral level, SimStudent functions as a *synthetic tutee*—an interactive machine learning agent that acquires cognitive skills through worked examples and guided problem-solving. From a technological perspective, it represents a novel realization of *programming by demonstration*, integrating multiple AI and machine learning techniques to simulate human-like learning.

The SimStudent technology has supported three major lines of inquiry, each contributing to a different facet of learning sciences and educational technology:

1. **Intelligent Authoring**—Investigating how non-programmers can build intelligent tutoring systems by tutoring the synthetic tutee.
2. **Student Modeling**—Using SimStudent to simulate human learning processes, enabling deeper insights into how students acquire complex knowledge.
3. **Learning by Teaching**—Exploring how and when students improve their own learning by teaching SimStudent as a synthetic peer to solve complex problems.

I initiated the SimStudent project in 2005 during my postdoctoral fellowship at Carnegie Mellon University. Since then, I have led the project and grown it into a multimillion-dollar research initiative. I have secured six major federal grants in this area, totaling over $6.3 million, serving as Principal Investigator on all but one for which I was a postdoctoral researcher.

To date, the SimStudent project has resulted in nine journal publications—including in high-impact venues such as *Artificial Intelligence*, *Journal of Educational Psychology*, and *Journal of Artificial Intelligence in Education*—and 38 peer-reviewed conference papers. The project continues to contribute both theoretical advancements and practical tools at the intersection of AI, cognitive science, and education.

Intelligent Authoring Aid with a Synthetic Peer

I have developed an innovative technology for authoring intelligent tutoring systems (ITSs) by interactively tutoring **SimStudent** on how to solve the target problems that the ITS will eventually teach to students (Matsuda, Cohen, & Koedinger, 2005; Matsuda, Cohen, Sewall, Lacerda, & Koedinger, 2008). In this context, SimStudent serves as a core component within the existing Cognitive Tutor Authoring Tools (CTAT) suite. A *Cognitive Tutor* is a type of ITS that supports mastery learning of problem-solving skills.

With CTAT, an author first designs a graphical user interface (GUI) for the desired cognitive tutor. The author then "tutors" SimStudent through this GUI. SimStudent learns a set of skills for the target problems, encoded as production rules, which then serve as the expert model for the cognitive tutor.

Empirical studies have demonstrated that using SimStudent for authoring cognitive tutors significantly streamlines the authoring process and supports wider dissemination of adaptive tutoring systems. One such study showed that this pedagogical agent technology has the potential to substantially reduce the time and expertise required to develop effective tutors (Matsuda, Cohen, & Koedinger, 2015).

Another study showed that SimStudent can support *cognitive task analysis*, a critical step in tutor development. Specifically, failures by SimStudent to solve target problems can highlight flaws in the cognitive model provided—such as missing background knowledge or misconfigured interface elements—thus helping authors refine the tutor design (Matsuda, 2022).

Student Modeling by Simulating Student Learning with a Pedagogical Agent

The SimStudent technology enables tightly controlled simulation studies that support the investigation of both domain-specific and domain-general theories of learning. One example of advancing a domain-specific theory involves my research on how students make *induction errors* in algebra that lead to the acquisition of incorrect skills (Matsuda, Epstein, Cohen, & Koedinger, 2008; Matsuda, Lee, Cohen, & Koedinger, 2009). I hypothesized that such errors stem from shallow, perceptually grounded prerequisite knowledge. This type of knowledge may cause students to manipulate algebraic symbols based on visual patterns rather than conceptual understanding—for example, interpreting the "3" in "3x" as simply a number followed by a letter, rather than as the coefficient of a variable term.

To test this hypothesis, I conducted a simulation study in which SimStudent was only given access to shallow prerequisite knowledge. The resulting behavior closely mirrored actual student learning patterns, including the common induction errors observed in classroom settings. The improved fit between SimStudent's behavior and empirical student data provided theoretical evidence for how conceptual prerequisites shape skill acquisition in algebra. These results demonstrate SimStudent's non-trivial potential to contribute to learning theory by enabling rigorous simulation-based experimentation.

Studying the Theory of Learning by Teaching with the Synthetic Peer

While prior research has shown that students can benefit from teaching others, the underlying cognitive mechanisms of this effect remain underexplored. To advance a theoretical understanding of *learning by teaching*, I launched a research program leveraging SimStudent as a synthetic peer, i.e., a *teachable agent*. I designed and developed an interactive, game-like online learning environment in which students learn to solve algebraic equations by teaching SimStudent.

To date, we have conducted eight evaluation studies involving more than 2,000 middle school students in real algebra classrooms. Key findings from this research program include: (a) Students improve their skills to solve questions by teaching SimStudent (Matsuda, Yarzebinski, Keiser, Raizada, Stylianides, et al., 2012). (b) The effectiveness of learning by teaching is significantly influenced by the student's prior knowledge (Matsuda et al., 2013). (c) Student learning is tightly coupled with the quality of the tutee's learning: when SimStudent engages in shallow learning, the corresponding student learning is also diminished (Matsuda, Yarzebinski, Keiser, Raizada, Cohen, et al., 2012). (d) Students who provide *elaborated answers* to SimStudent's questions demonstrate greater learning gains (Matsuda, Cohen, et al., 2012; Shahriar & Matsuda, 2023, 2024). (e) Learning by teaching is most effective when supported by *adaptive scaffolding*, particularly scaffolding that guides *how to teach* (as opposed to how to solve) the target task (Matsuda, Lv, & Zheng, 2023; Matsuda, Weng, & Wall, 2020).

A key methodological strength of this work lies in the use of SimStudent to conduct *tightly controlled studies* in authentic classroom settings while capturing both detailed learning interaction data and learning outcome data. This enables the investigation of fine-grained cognitive and metacognitive processes in ecologically valid learning environments.

1.2 Evidence-based Learning-Engineering Methods for Building Online Courseware at Scale

This line of research focuses on developing *evidence-based, transformative engineering methods* for the efficient creation of *adaptive online courseware*—in the spirit of platforms such as Coursera and edX, but with a stronger emphasis on adaptive pedagogy. The central hypothesis is that analyzing large-scale *student learning log data* (e.g., performance records, assessment outcomes) in conjunction with *instructional content data* (e.g., text instructions, video lectures) can yield powerful insights into the design of more effective and personalized online learning experiences. This research aims to bridge the gap between learning analytics and instructional design to create courseware that dynamically adapts to individual learners' needs, thereby enhancing both learning efficiency and scalability.

Data-driven Online Courseware Development Environment

To support the evidence-based design of online courseware, I have developed an integrated authoring platform for the data-driven creation and iterative refinement of online learning materials, called **IDEA** (Integrated Development Environment with learning Analytics). The conceptual foundation for IDEA emerged from contextual interviews conducted with experienced online courseware developers during my tenure at Carnegie Mellon University. These interviews informed the development of features tailored to the real-world needs of instructional designers.

IDEA provides courseware developers with *real-time, data-driven feedback* on course structure—for example, identifying gaps such as the absence of formative assessments linked to specific learning objectives. The system is designed as a visually interactive, WYSIWYG (What You See Is What You Get) authoring tool, enabling developers to receive actionable insights during the courseware creation process.

We deployed IDEA on the Open Learning Initiative (OLI) platform and applied it to the iterative redesign of several online courses. An observatory evaluation of an IDEA-enhanced *Discrete Mathematics* course—used as a gateway course for first-year undergraduate students in the School of Computer Science at Carnegie Mellon University—indicated both reduced learning time and improved learning outcomes.

This project provides empirical support for the feasibility and effectiveness of a *data-driven, iterative approach to online courseware engineering*, which we argue is a foundational strategy for scaling high-quality, adaptive online education.

Scalable Learning Engineering Methods

More recently, I launched the **PASTEL** project (*Pragmatic Methods to Develop Adaptive and Scalable Technologies for Next-Generation E-Learning*) to advance *learning-engineering methods* for building adaptive online courseware at scale. The primary goal of PASTEL is to investigate evidence-based, transformative engineering practices that enable instructional designers and engineers to efficiently create *CyberBook*—a novel integration of cognitive tutors within online courseware environments.

CyberBook merges the adaptive instructional power of cognitive tutors (which support mastery learning through personalized scaffolding) with the multi-modal, media-rich content delivery of traditional online platforms. I hypothesize that this integration will promote *synergetic competency*—robust learning characterized by a strong concurrent (or "bidirectional," if you will) growth of *conceptual understanding* and *procedural fluency*.

The PASTEL project is currently focused on developing and evaluating six core technologies: (1) A **text-mining method** to uncover latent concepts and skills embedded in didactic instructional text. (2) A **web-based authoring tool** that allows developers to demonstrate problem solutions and seamlessly generate intelligent tutoring components. (3) A **prompt engineering technique** for large language models to generate assessment questions aligned with learning objectives and student common misconceptions. (4) A **reinforcement learning framework** to sequence course content optimally based on individual student proficiency. (5) A **stochastic prediction algorithm** for early identification of at-risk students. (6) A **quality-assurance technique** that leverages student performance data using reinforcement learning to assess and refine courseware content.

These technologies are designed to empower courseware developers with data-driven, AI-powered tools for building adaptive learning experiences efficiently. The ultimate aim is to achieve scalable, high-quality online education that can serve diverse learners around the world.

Those six technologies have been prototyped and integrated into Open Learning Initiative (OLI) at CMU and Open edX as example platforms (Matsuda, Shimmei, et al., 2023). Preliminary findings from empirical studies include: (a) The stochastic prediction algorithm for risk detection demonstrated high recall (0.79), successfully identifying most students truly at risk, though with modest precision (0.25), indicating room for refinement in reducing false positives (Matsuda, Chandrasekaran, & Stamper, 2016). (b) The text-mining method for latent skill discovery outperformed a human-authored model in predicting learning outcomes in the OLI Biology course (Matsuda, Furukawa, Bier, & Faloutsos, 2015). (c) A pre-trained large language model (OpenAI GPT) can generate analytical multiple-choice questions that resemble the item characteristics (estimated by the Item Response Theory) of those crafted by experienced instructors when students' written responses to open-ended questions were given and instructed to analyze them to identify notable misconceptions (Shimmei et al., 2025; *the Best Late Breaking Results Paper*).

The PASTEL project also contributes to advancing *foundational AI theories*. For instance, in our work on applying large language models for automated question generation, we proposed an innovative ensemble learning technique designed to address challenges associated with training deep learning models with small data regimes (Shimmei & Matsuda, 2023). This work was recognized with an *Honorable Mentor Award* at the International Conference on Educational Data Mining, underscoring both its technical novelty and scholarly impact.

Beyond these technical contributions, the broader impact of the PASTEL project lies in its potential to support global dissemination of a *next-generation cyberlearning infrastructure*—enabling researchers and educators to collaboratively explore and deploy scalable, adaptive, evidence-based learning environments.

2. Future Directions

Building on my prior work, I will continue to advance the theory of learning and teaching by leveraging *technological innovation* as a driving force for transformation. Today's education systems face complex, multifaceted challenges that demand solutions grounded in a deep understanding of students', teachers', and practitioners' real-world needs. My research has consistently demonstrated the power of user-centered, iterative development—guided by both technological rigor and theoretical insight—to address such challenges effectively.

Going forward, I will continue to explore the *theory and design of intelligent pedagogical agents* and *adaptive online courseware engineering* for a wide range of learning contexts, including K-16 education, lifelong learning, and workforce training. These applications will support my broader research agenda: to harness artificial intelligence in education in ways that enable personalized, scalable, and accessible learning for diverse populations. Ultimately, my long-term goal is to contribute to the *broad dissemination* of research-driven educational technologies that promote effective learning and equitable access to quality education across social, geographic, and economic boundaries.

References:

Matsuda, N. (2022). Teachable Agent as an Interactive Tool for Cognitive Task Analysis: A Case Study for Authoring an Expert Model. *International Journal of Artificial Intelligence in Education, 32*, 48-75. doi: 10.1007/s40593-021-00265-z

Matsuda, N., Chandrasekaran, S., & Stamper, J. (2016). How quickly can wheel spinning be detected? In T. Barnes, M. Chi & M. Feng (Eds.), *Proceedings of the International Conference on Educational Data Mining* (pp. 607-608).

Matsuda, N., Cohen, W. W., & Koedinger, K. R. (2005). Applying Programming by Demonstration in an Intelligent Authoring Tool for Cognitive Tutors *AAAI Workshop on Human Comprehensible Machine Learning (Technical Report WS-05-04)* (pp. 1-8). Menlo Park, CA: AAAI association.

Matsuda, N., Cohen, W. W., & Koedinger, K. R. (2015). Teaching the Teacher: Tutoring SimStudent leads to more Effective Cognitive Tutor Authoring. *International Journal of Artificial Intelligence in Education, 25*, 1-34. doi: 10.1007/s40593-014-0020-1

Matsuda, N., Cohen, W. W., Koedinger, K. R., Keiser, V., Raizada, R., Yarzebinski, E., . . . Stylianides, G. J. (2012). Studying the Effect of Tutor Learning using a Teachable Agent that asks the Student Tutor for Explanations. In M. Sugimoto, V. Aleven, Y. S. Chee & B. F. Manjon (Eds.), *Proceedings of the International Conference on Digital Game and Intelligent Toy Enhanced Learning (DIGITEL 2012)* (pp. 25-32). Los Alamitos, CA: IEEE Computer Society.

Matsuda, N., Cohen, W. W., Sewall, J., Lacerda, G., & Koedinger, K. R. (2008). Why Tutored Problem Solving may be better than Example Study: Theoretical Implications from a Simulated-Student Study. In B. P. Woolf, E. Aimeur, R. Nkambou & S. Lajoie (Eds.), *Proceedings of the International Conference on Intelligent Tutoring Systems* (pp. 111-121). Heidelberg, Berlin: Springer.

Matsuda, N., Epstein, S., Cohen, W. W., & Koedinger, K. R. (2008). Towards a theory of learning errors: application of a synthetic student to analyze students errors *Proceedings of Japan National Conference on Information and Systems in Education* (pp. 362-363). Kumamoto, Japan.

Matsuda, N., Furukawa, T., Bier, N., & Faloutsos, C. (2015). Machine beats experts: Automatic discovery of skill models for data-driven online course refinement. In J. G. Boticario, O. C. Santos, C. Romero, M. Pechenizkiy, A. Merceron, P. Mitros, J. M. Luna, C. Michaescu, P. Moreno, A. Hershkovitz, S. Ventura & M. C. Desmarais (Eds.), *Proceedings of the International Conference on Educational Data Mining* (pp. 101-108). Madrid, Spain.

Matsuda, N., Lee, A., Cohen, W. W., & Koedinger, K. R. (2009). A Computational Model of How Learner Errors Arise from Weak Prior Knowledge. In N. Taatgen & H. van Rijn (Eds.), *Proceedings of the Annual Conference of the Cognitive Science Society* (pp. 1288-1293). Austin, TX: Cognitive Science Society.

Matsuda, N., Lv, D., & Zheng, G. (2023). Teaching How to Teach Promotes Learning by Teaching. *International Journal of Artificial Intelligence in Education, 33*, 720–751. doi: 10.1007/s40593-022-00306-1

Matsuda, N., Shimmei, M., Chaudhuri, P., Makam, D., Shrivastava, R., Wood, J., & Taneja, P. (2023). PASTEL: Evidence-based learning engineering methods to facilitate creation of adaptive online courseware. In F. Ouyang, P. Jiao, B. M. McLaren & A. H. Alavi (Eds.), *Artificial Intelligence in STEM Education: The Paradigmatic Shifts in Research, Education, and Technology* (pp. 93-108). New York, NY: CSC Press.

Matsuda, N., Weng, W., & Wall, N. (2020). The effect of metacognitive scaffolding for learning by teaching a teachable agent. *International Journal of Artificial Intelligence in Education, 30*(1), 1-37.

Matsuda, N., Yarzebinski, E., Keiser, V., Raizada, R., Cohen, W. W., Stylianides, G. J., & Koedinge, K. R. (2012). Shallow learning as a pathway for successful learning both for tutors and tutees. In N. Miyake, D. Peebles & R. P. Cooper (Eds.), *Proceedings of the 34th Annual Conference of the Cognitive Science Society* (pp. 731-736). Austin, TX: Cognitive Science Society.

Matsuda, N., Yarzebinski, E., Keiser, V., Raizada, R., Stylianides, G. J., & Koedinger, K. R. (2012). Motivational factors for learning by teaching: The effect of a competitive game show in a virtual peer-learning environment. In S. Cerri & W. Clancey (Eds.), *Proceedings of International Conference on Intelligent Tutoring Systems* (pp. 101-111). Heidelberg, Berlin: Springer-Verlag.

Matsuda, N., Yarzebinski, E., Keiser, V., Raizada, R., William, W. C., Stylianides, G. J., & Koedinge, K. R. (2013). Cognitive anatomy of tutor learning: Lessons learned with SimStudent. *Journal of Educational Psychology, 105*(4), 1152-1163. doi: 10.1037/a0031955

Shahriar, T., & Matsuda, N. (2023). What and how you explain matters: Inquisitive Teachable Agent Scaffolds Knowledge-building for Tutor Learning. In N. Wang, G. Rebolledo-Mendez, O. C. Santos, V. Dimitrova & N.

Matsuda (Eds.), *Proceedings of the International Conference on Artificial Intelligence in Education* (pp. 126-138): Springer.

Shahriar, T., & Matsuda, N. (2024). "I am confused! How to differentiate between…?" Adaptive Followup Questions Facilitate Tutor Learning with Effective Time-on-task. In Andrew, Irene & Zitao (Eds.), *Proceedings of the International Conference on Artificial Intelligence in Education* (pp. 1-14): Springer.

Shimmei, M., & Matsuda, N. (2023). Can't Inflate Data? Let the Models Unite and Vote: Data-agnostic Method to Avoid Overfit with Small Data. In R. Agrawal, Y. Narahari, M. Pechenizkiy, M. Feng, T. Käser & P. Talukdar (Eds.), *Proceedings of the International Conference on Educational Data Mining* (pp. 286-295): Educational Data Mining Society.

Shimmei, M., Uto, M., Matsubayashi, Y., Inui, K., Mallavarapu, A., & Matsuda, N. (2025). Tell me who your students are: GPT can generate valid multiple-choice questions when students' (mis)understanding is hinted. In A. I. Cristea, E. Walker, L. Yu, O. C. Santos & S. Isotani (Eds.), *Proceedings of the International Conference on Artificial Intelligence in Education* (pp. 126-134; Late Breaking Results): Springer.